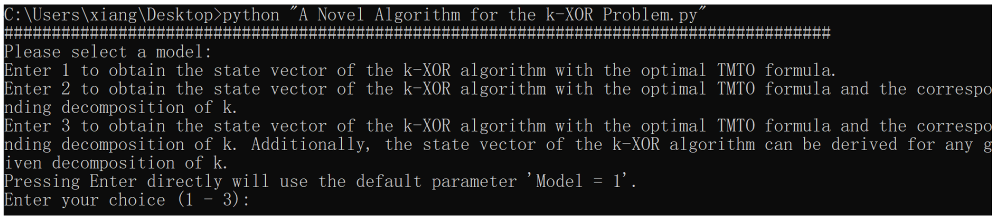
Design the k-XOR algorithm with the optimal TMTO formula based on our code.

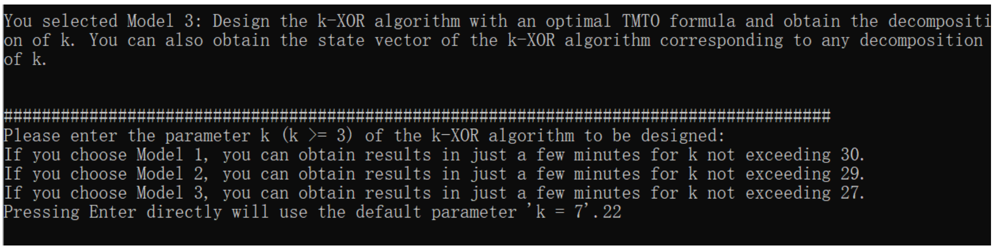
Step-1. Run the code "`A Novel Algorithm for the k-XOR Problem.py`" and type `1, 2, or 3` to select the corresponding model.
- If you choose `Model 1`, you can obtain results in just a few minutes for k not exceeding 30.
- If you choose `Model 2`, you can obtain results in just a few minutes for k not exceeding 29.
- If you choose `Model 3`, you can obtain results in just a few minutes for k not exceeding 26.



```
C:\Users\xiang\Desktop>python "A Novel Algorithm for the k-XOR Problem.py"
##########################################################################################
Please select a model:
Enter 1 to obtain the state vector of the k-XOR algorithm with the optimal TMTO formula.
Enter 2 to obtain the state vector of the k-XOR algorithm with the optimal TMTO formula and the correspo
nding decomposition of k.
Enter 3 to obtain the state vector of the k-XOR algorithm with the optimal TMTO formula and the correspo
nding decomposition of k. Additionally, the state vector of the k-XOR algorithm can be derived for any g
iven decomposition of k.
Pressing Enter directly will use the default parameter 'Model = 1'.
Enter your choice (1 - 3):
```

Fig. 7 Step-1.

Step-2. If you typed `3` in the Step-1, you need to continue to enter the parameter k to determine the k-XOR algorithm you want to design.



```
You selected Model 3: Design the k-XOR algorithm with an optimal TMTO formula and obtain the decompositi
on of k. You can also obtain the state vector of the k-XOR algorithm corresponding to any decomposition
of k.


##########################################################################################
Please enter the parameter k (k >= 3) of the k-XOR algorithm to be designed:
If you choose Model 1, you can obtain results in just a few minutes for k not exceeding 30.
If you choose Model 2, you can obtain results in just a few minutes for k not exceeding 29.
If you choose Model 3, you can obtain results in just a few minutes for k not exceeding 27.
Pressing Enter directly will use the default parameter 'k = 7'.22
```

Fig. 8 Step-2.

Step-3. Suppose we need to design a 22-XOR algorithm with the optimal TMTO formula.

After entering 22, we can obtain the corresponding state vector of the 22-XOR algorithm with the optimal TMTO formula:

$$[[4,0,1],[5,1,1],[5,2,1],[6,3,1],[6,4,1],[6,5,1],[6,6,1],[6,7,1],[6,8,1],[6,9,1],[6,10,1],[6,11,1],[6,12,1],[6,13,1],[6,14,1],[6,15,1]] \tag{11}$$

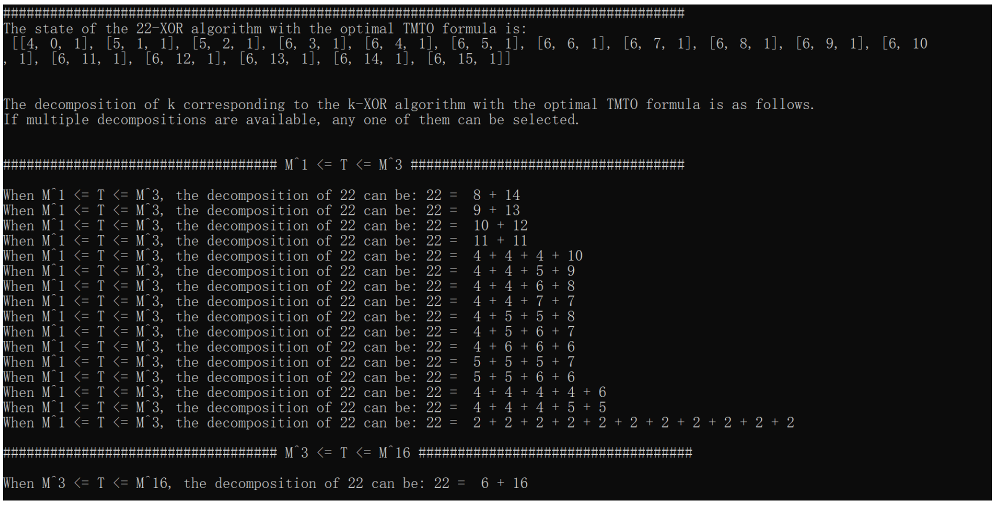
The state vector is consistent with Table B1 in the manuscript. According to Definition 1 in the manuscript, we can calculate the matching lengths of the subroot list and the root list of the 22-XOR algorithm under varying time complexities, based on the aforementioned state vector, as demonstrated in Table 2.

Table 2 The matching lengths of the subroot list and the root list of the 22-xor algorithm with the optimal TMTO formula.

The time complexity	2^m	2^{2m}	2^{3m}	2^{4m}
The matching length of the subroot list	$(4+0)m$	$(5+1)m$	$(5+2)m$	$(6+3)m$
The matching lengths of the root list	$(4+0+1)m$	$(5+1+1)m$	$(5+2+1)m$	$(6+3+1)m$

Similarly, we can calculate the matching lengths of the subroot list and the root list of the 22-XOR algorithm when the time complexity is 2^{tm} $(5 \leq t \leq 16)$, though for brevity, we refrain from providing the detailed list here. Subsequently, we can derive the optimal TMTO formula (Equation 12) for the 22-XOR algorithm according to Table 2. The result is consistent with Table B4 in the manuscript.

$$N = \begin{cases} T^2M^3, & M \leq T \leq M^2, \\ TM^5, & M^2 \leq T \leq M^3, \\ T^2M^2, & M^3 \leq T \leq M^4, \\ N = TM^6 & M^4 \leq T \leq M^{16}. \end{cases} \tag{12}$$



```
#######################################################################################
The state of the 22-XOR algorithm with the optimal TMTO formula is:
 [[4, 0, 1], [5, 1, 1], [5, 2, 1], [6, 3, 1], [6, 4, 1], [6, 5, 1], [6, 6, 1], [6, 7, 1], [6, 8, 1], [6, 9, 1], [6, 10
, 1], [6, 11, 1], [6, 12, 1], [6, 13, 1], [6, 14, 1], [6, 15, 1]]


The decomposition of k corresponding to the k-XOR algorithm with the optimal TMTO formula is as follows.
If multiple decompositions are available, any one of them can be selected.


################################### M^1 <= T <= M^3 ###################################

When M^1 <= T <= M^3, the decomposition of 22 can be: 22 =  8 + 14
When M^1 <= T <= M^3, the decomposition of 22 can be: 22 =  9 + 13
When M^1 <= T <= M^3, the decomposition of 22 can be: 22 =  10 + 12
When M^1 <= T <= M^3, the decomposition of 22 can be: 22 =  11 + 11
When M^1 <= T <= M^3, the decomposition of 22 can be: 22 =  4 + 4 + 4 + 10
When M^1 <= T <= M^3, the decomposition of 22 can be: 22 =  4 + 4 + 5 + 9
When M^1 <= T <= M^3, the decomposition of 22 can be: 22 =  4 + 4 + 6 + 8
When M^1 <= T <= M^3, the decomposition of 22 can be: 22 =  4 + 4 + 7 + 7
When M^1 <= T <= M^3, the decomposition of 22 can be: 22 =  4 + 5 + 5 + 8
When M^1 <= T <= M^3, the decomposition of 22 can be: 22 =  4 + 5 + 6 + 7
When M^1 <= T <= M^3, the decomposition of 22 can be: 22 =  4 + 6 + 6 + 6
When M^1 <= T <= M^3, the decomposition of 22 can be: 22 =  5 + 5 + 5 + 7
When M^1 <= T <= M^3, the decomposition of 22 can be: 22 =  5 + 5 + 6 + 6
When M^1 <= T <= M^3, the decomposition of 22 can be: 22 =  4 + 4 + 4 + 4 + 6
When M^1 <= T <= M^3, the decomposition of 22 can be: 22 =  4 + 4 + 4 + 5 + 5
When M^1 <= T <= M^3, the decomposition of 22 can be: 22 =  2 + 2 + 2 + 2 + 2 + 2 + 2 + 2 + 2 + 2 + 2

################################### M^3 <= T <= M^16 ###################################

When M^3 <= T <= M^16, the decomposition of 22 can be: 22 =  6 + 16
```

Fig. 9 Step-3.

At the same time, the code also returns all possible decompositions corresponding to the 22-XOR algorithm with the optimal TMTO formula:

When $M^1 <= T <= M^3$, the decomposition of 22 can be: $22 = 8 + 14$.
When $M^1 <= T <= M^3$, the decomposition of 22 can be: $22 = 9 + 13$.
When $M^1 <= T <= M^3$, the decomposition of 22 can be: $22 = 10 + 12$.
When $M^1 <= T <= M^3$, the decomposition of 22 can be: $22 = 11 + 11$.
When $M^1 <= T <= M^3$, the decomposition of 22 can be: $22 = 4 + 4 + 4 + 10$.
When $M^1 <= T <= M^3$, the decomposition of 22 can be: $22 = 4 + 4 + 5 + 9$.
When $M^1 <= T <= M^3$, the decomposition of 22 can be: $22 = 4 + 4 + 6 + 8$.
When $M^1 <= T <= M^3$, the decomposition of 22 can be: $22 = 4 + 4 + 7 + 7$.
When $M^1 <= T <= M^3$, the decomposition of 22 can be: $22 = 4 + 5 + 5 + 8$.
When $M^1 <= T <= M^3$, the decomposition of 22 can be: $22 = 4 + 5 + 6 + 7$.
When $M^1 <= T <= M^3$, the decomposition of 22 can be: $22 = 4 + 6 + 6 + 6$.
When $M^1 <= T <= M^3$, the decomposition of 22 can be: $22 = 5 + 5 + 5 + 7$.
When $M^1 <= T <= M^3$, the decomposition of 22 can be: $22 = 5 + 5 + 6 + 6$.
When $M^1 <= T <= M^3$, the decomposition of 22 can be: $22 = 4 + 4 + 4 + 4 + 6$.
When $M^1 <= T <= M^3$, the decomposition of 22 can be: $22 = 4 + 4 + 4 + 5 + 5$.
When $M^1 <= T <= M^3$, the decomposition of 22 can be: $22 = 2 + 2 + 2 + 2 + 2 + 2 + 2 + 2 + 2 + 2 + 2$.
When $M^3 <= T <= M^{16}$, the decomposition of 22 can be: $22 = 6+16$.

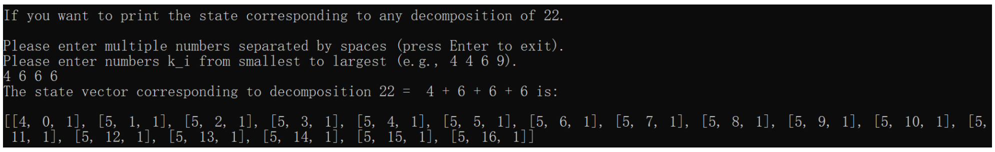
It should be emphasized that when $M^1 <= T <= M^3$, there are multiple decompositions of k, and we can choose any one. The above results are consistent with Table B4 in the manuscript.

Step-4. Finally, if we wish to obtain the state vector of the 22-XOR algorithm corresponding to a specific decomposition of 22, please type the decomposition of 22 according to the following rules:
`- Please enter multiple numbers separated by spaces.`
`- Please enter numbers` k_i `from smallest to largest (e.g., 4 4 6 9)` ($k = \sum_{0 \leq i \leq r-1} k_i$).

Taking the decomposition of $22 = 4 + 6 + 6 + 6$ as an example, the code will immediately return the state vector of the 22-XOR algorithm corresponding to the decomposition of $22 = 4 + 6 + 6 + 6$. That is:

$$
\begin{aligned}
&[[4,0,1],[5,1,1],[5,2,1],[5,3,1],[5,4,1],[5,5,1],[5,6,1],[5,7,1],[5,8,1],[5,9,1],\\
&[5,10,1],[5,11,1],[5,12,1],[5,13,1],[5,14,1],[5,15,1],[5,16,1]]
\end{aligned}
\tag{13}
$$

By comparing the state vectors corresponding to Equations 11 and 13, we can find that the 22-XOR algorithm corresponding to decomposition $22 = 4 + 6 + 6 + 6$ is not the 22-XOR algorithm with the optimal TMTO formula.



```
If you want to print the state corresponding to any decomposition of 22.

Please enter multiple numbers separated by spaces (press Enter to exit).
Please enter numbers k_i from smallest to largest (e.g., 4 4 6 9).
4 6 6 6
The state vector corresponding to decomposition 22 =  4 + 6 + 6 + 6 is:

[[4, 0, 1], [5, 1, 1], [5, 2, 1], [5, 3, 1], [5, 4, 1], [5, 5, 1], [5, 6, 1], [5, 7, 1], [5, 8, 1], [5, 9, 1], [5, 10, 1], [5,
 11, 1], [5, 12, 1], [5, 13, 1], [5, 14, 1], [5, 15, 1], [5, 16, 1]]
```

Fig. 10 Step-4.

Step-5. Deriving the TMTO formula of the $\bar{k}$-XOR algorithm based on the TMTO formula of the k-XOR algorithm.

After obtaining the TMTO formula of the k-XOR algorithm, we can choose the preparation algorithm and derive the TMTO formula for the $\bar{k}$-XOR algorithm. As an example, we illustrate the derivation of the TMTO formula for the 14-XOR algorithm based on the TMTO formula of the 7-XOR algorithm.

Assuming that the TMTO formulas of the k-XOR algorithm are

$$N = \begin{cases} T^{\beta_0} M^{\gamma_0}, & M^{\zeta_0} \le T \le M^{\delta_0}; \\ T^{\beta_1} M^{\gamma_1}, & M^{\zeta_1} \le T \le M^{\delta_1}; \\ \cdots, & \cdots; \\ T^{\beta_{v-1}} M^{\gamma_{v-1}}, & M^{\zeta_{v-1}} \le T \le M^{\delta_{v-1}}. \end{cases}$$

Example 6. *The TMTO formulas of the 7-XOR algorithm are*

$$N = \begin{cases} T^2 M, & M \le T \le M^2; \\ TM^3, & M^2 \le T \le M^4, \end{cases}$$

and we have $\beta_0 = 2$, $\gamma_0 = 1$, $\delta_0 = 2$ *and* $\beta_1 = 1$, $\gamma_1 = 3$, $\delta_1 = 4$.

Let $L = 2^l$, then the TMTO formulas of $\bar{k}$-XOR algorithm are shown in Table 3. PCS, PCSIEF, and CTP represent the selected preparation algorithm.

Table 3 The TMTO formulas of the $\bar{k}$-XOR alogrithm.

No.	TMTO Formula	k	Range (T vs. M)	Range (L)	Building Blocks
1	$N = T^{\beta+2} M^{\gamma-1}$	$\bar{k}/2$	$T \le M^{\delta}$	$l \ge \frac{2n-2m\gamma-m\beta}{\beta+2}$	PCS + $A_{\bar{k}/2}$
2	$N = T^2 M^{\delta\beta+\gamma-1}$	$\bar{k}/2$	$T \ge M^{\delta}$	$l \ge n - m(\delta\beta+\gamma)$	PCS + $A_{\bar{k}/2}$
3	$N = T^{\beta+1} M^{\gamma-\frac{1}{2}} L^{\frac{1}{2}}$	$\bar{k}/2$	$T \le M^{\delta}$	$l \le \frac{2n-2m\gamma-m\beta}{\beta+2}$	PCSIEF + $A_{\bar{k}/2}$
4	$N = TM^{\delta\beta+\gamma-1} L^{\frac{1}{2}}$	$\bar{k}/2$	$T \ge M^{\delta}$	$l \le n - m(\delta\beta+\gamma)$	PCSIEF + $A_{\bar{k}/2}$
5	$N = T^{\beta+1} M^{\gamma-1}$	$\bar{k}$	$T \le M^{\delta}$	$l \ge \frac{n-m\gamma+m}{\beta+1}$	CTP + $A_{\bar{k}}$
6	$N = T^{\beta} M^{\gamma-1} L$	$\bar{k}$	$T \le M^{\delta}$	$l \le \frac{n-m\gamma+m}{\beta+1}$	CTP + $A_{\bar{k}}$

Example 7. *When* $M^2 \le T \le M^4$, *the TMTO formula of the 7-XOR algorithm is* $N = TM^3$, *that is* $\beta = 1$, $\gamma = 3$, $\delta = 4$. *If* $l \le \frac{2n-7m}{3}$, *we can derive the (partial) TMTO formula of the 14-XOR algorithm from row 4 (The third formula) of Table 3 as*

$$N = T^2 M^{\frac{5}{2}} L^{\frac{1}{2}}.$$